

23. Element Size aur Scrolling

Ye properties elements ki width, height aur position ke bare me bataati hain. Sab values **pixels me numbers** hoti hain (string nahi).

Sample element

`width: 300px; height: 200px; border: 25px; padding: 20px; overflow: auto` (scrollbar ke saath). **Margin ka koi property nahi** hota (wo element ka hissa nahi).

Dhyan: kuch browsers scrollbar ki jagah content se lete hain (jaise 16px), isliye content width `300 - 16 = 284px` ho sakti hai.

Properties (bahar se andar)

Property	Matlab	Sample value
<code>offsetParent</code>	sabse paas ka positioned ancestor (ya <code>td/th/table/body</code>)	
<code>offsetLeft / offsetTop</code>	<code>offsetParent</code> ke upper-left corner se coordinates	
<code>offsetWidth / offsetHeight</code>	outer size , borders ke saath	390 / 290
<code>clientLeft / clientTop</code>	borders ki width (asal me outer corner se inner corner ki doori)	25 / 25
<code>clientWidth / clientHeight</code>	andar ka area: content + padding, scrollbar aur border nahi	324 / 240
<code>scrollWidth / scrollHeight</code>	client jaisa, par chhupa (scroll hua) hissa bhi	324 / 723
<code>scrollLeft / scrollTop</code>	kitna hissa scroll ho chuka (upar/left chhupa)	

Kuch important baatein

- Ye sab **read-only** hain, sirf `scrollLeft/scrollTop` **badal sakte ho** (element scroll ho jaayega).
 - `elem.scrollTop += 10` (10px neeche)
 - `scrollTop = 0` (sabse upar), `scrollTop = 1e9` (sabse neeche)
- Hidden elements** (`display:none` ya document me nahi): saari geometry properties `0` aur `offsetParent` `null`.

js

```
function isHidden(elem) { return !elem.offsetWidth && !elem.offsetHeight; }
```

- `offsetParent` `null` होता है: hidden elements, `body/html`, और `position:fixed` elements के लिए.
- Pura content dikhane के लिए: `element.style.height = element.scrollHeight + 'px'`

CSS `width/height` kyun na lo (`getComputedStyle`)?

1. CSS width `box-sizing` par depend karti hai.
2. CSS width `auto` ho sakti hai (inline elements), JS ko exact `px` chahiye.
3. Scrollbar ka behavior browsers me alag (Chrome scrollbar minus karke deta hai, Firefox CSS width). `clientWidth` in sab me consistent hai.

Tasks ke jawab (short)

- Neeche se kitna scroll baaki? `scrollTop = elem.scrollHeight - elem.scrollTop - elem.clientHeight`
- Scrollbar ki width: ek `overflow-y: scroll` wala div banao (document me daalo), `offsetWidth - clientWidth`.
- Ball ko field ke center me:

js

```
ball.style.left = Math.round(field.clientWidth / 2 - ball.offsetWidth / 2) + 'p  
ball.style.top = Math.round(field.clientHeight / 2 - ball.offsetHeight / 2) +
```

Pitfall: `<img>` ki width/height nahi di to load hone tak `offsetWidth = 0` milta hai, isliye HTML/CSS me size do.

- CSS width vs `clientWidth`: `clientWidth` number hai (string nahi), `auto` nahi aata, padding include karta hai, scrollbar hamesha sahi handle karta hai.

Yaad rakho

`offset*` = outer (border के साथ), `client*` = andar (border aur scrollbar के बina), `scroll*` = पूरा content (chhupa hissa भी).